

30. A convex lens of focal length 15 cm, is forming a real image. If the size of image is same as the size of object, then position of object and position of image will be, respectively :

1

- (A) – 15 cm and – 15 cm from lens
- (B) – 15 cm and + 15 cm from lens
- (C) – 30 cm and + 30 cm from lens
- (D) – 30 cm and – 30 cm from lens

30. A convex lens of focal length 15 cm, is forming a real image. If the size of image is same as the size of object, then position of object and position of image will be, respectively :

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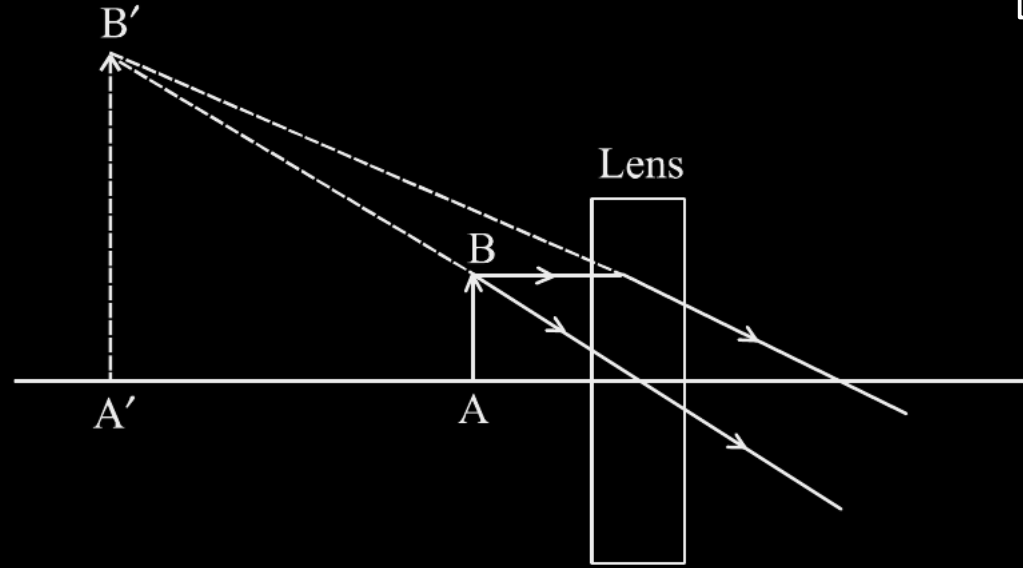
- (A) -15 cm and -15 cm from lens
- (B) -15 cm and $+15$ cm from lens
- (C) -30 cm and $+30$ cm from lens
- (D) -30 cm and -30 cm from lens

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(C) / -30 cm and $+30$ cm from lens

33. The given figure shows image formation by a lens. Analyse the figure and answer the following questions :

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- (a) What is the type of lens used for image formation in the given ray diagram ?
- (b) If the real image is formed at a distance of 30 cm from the lens and the size of image is twice the size of the object, then where was the object placed ?

(a)	Convex lens / Converging lens	1
(b)	m = - 2, v= 30 cm m = $\frac{v}{u}$ u = $\frac{v}{m}$ u = $\frac{30}{-2}$ u = -15cm / The object was placed at 15 cm in front of the lens.	$\frac{1}{2}$

38. Read the following passage and answer the questions given :

Lenses can form different types of images depending upon their focal length and position of object. A convex lens can create real, inverted or virtual, erect images, while a concave lens forms only virtual and diminished images. The focal length determines the power of lens. Convex lenses have positive focal length while concave lenses have negative focal length by convention. When lenses are placed together, their combined power is determined by the sum of their individual powers. Ray diagrams help to visualize how light converges or diverges through lens to form an image.

- (a) A convex lens of focal length 20 cm is used to form an image. If an object is placed at 40 cm from the lens, what will be the position and nature of image ?
- (b) Illustrate the formation of image with the help of ray diagram, when the object is placed between the optical centre and principal focus of concave lens.

(a)

- Position - Image will form at 40 cm / 2F / C
- Nature – Real and inverted

/

Alternate answer

$f = +20$ cm, $u = -40$ cm

Using Lens Formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-40} = \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{-40}$$

$v = +40$ cm

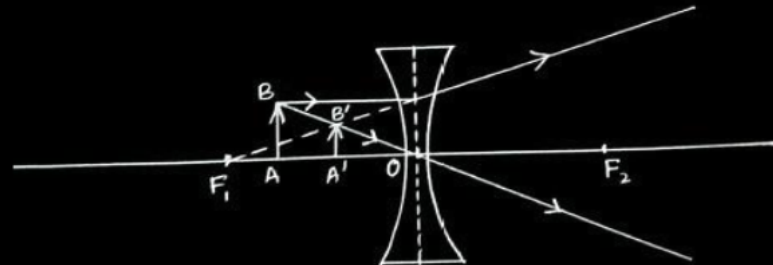
Position - 40 cm on the other side of the lens

Nature - Real and inverted

$\frac{1}{2}$

$\frac{1}{2}$

(b)



1

- (c) (i) A lens combination consists of a convex lens of focal length 30 cm and a concave lens of focal length 15 cm placed together. Find the equivalent focal length and power of this lens combination.

2

OR

- (c) (ii) Two lenses are placed in contact. One is a concave lens with focal length 2 m and the other is a convex lens with focal length 1.5 m. What type of lens will the combination behave as (convex or concave) ? Give reason.

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2

(c) (i)

$$f_1 = 30 \text{ cm} = 0.3 \text{ m}, \quad f_2 = -15 \text{ cm} = -0.15 \text{ m}$$

$$P = \frac{1}{f}$$

$$P_1 = \frac{+1}{0.3} \text{ D}; \quad P_2 = \frac{-1}{0.15} \text{ D}$$

Equivalent power, $P = P_1 + P_2$

$$P = -3.33 \text{ D}$$

Equivalent focal length, $f = \frac{1}{P}$

$$f = \frac{-1}{3.33} = -0.3 \text{ m} = -30 \text{ cm}$$

OR

(c) (ii)

- Combination Lens will behave like convex lens
- $f_1 = -2 \text{ m}, \quad f_2 = 1.5 \text{ m}$

$$P = \frac{1}{f}$$

$$P_1 = \frac{-1}{2} \text{ D}, \quad P_2 = \frac{+1}{1.5} \text{ D}$$

$$P = P_1 + P_2$$

$$P = \frac{1}{6}$$

$$f = +6 \text{ m}$$

The focal length of combination is positive.

/

Alternate answer

- Combination Lens will behave like convex lens.
- Convex lens - Less 'f', More 'P'
Concave lens - More 'f', Less 'P'
Combined Power = $P_1 + P_2$, which will be positive.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

1

34. A mirror always forms a virtual, erect and diminished image. Identify the mirror and draw a labelled ray diagram for image formation by this mirror.

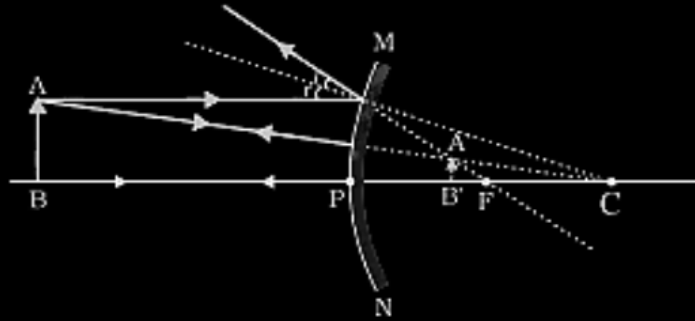
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34. A mirror always forms a virtual, erect and diminished image. Identify the mirror and draw a labelled ray diagram for image formation by this mirror.

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2

- Convex mirror / Diverging mirror
-



($\frac{1}{2}$ mark to be deducted for not showing the direction of light)

1

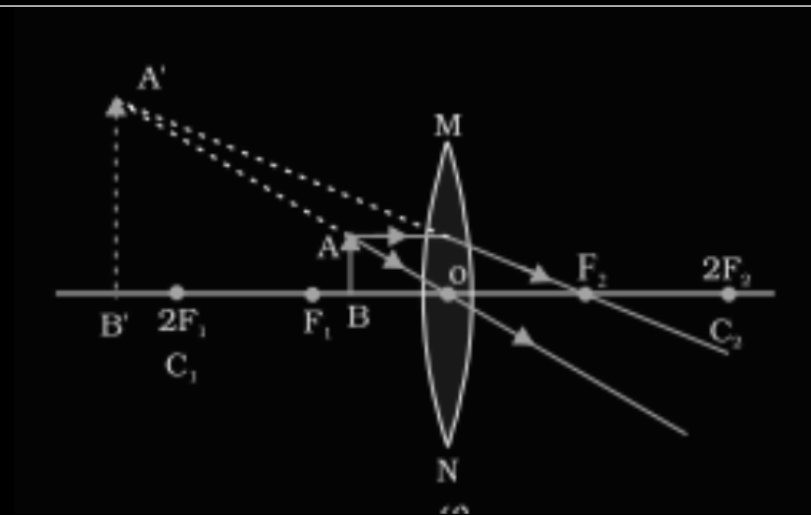
1

33. Draw the ray diagram for the image formation by a lens which shows a magnification of +2.

33. Draw the ray diagram for the image formation by a lens which shows a magnification of +2.

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2



(size of the image is approximate)

($\frac{1}{2}$ mark to be deducted for not showing the direction of light)

2

30. From the following options choose the one which gives negative magnification greater than one :

1

- (A) Concave mirror; Nature of image – real
- (B) Concave mirror; Nature of image – virtual
- (C) Convex mirror; Nature of image – real
- (D) Convex mirror; Nature of image – virtual

30. From the following options choose the one which gives negative magnification greater than one :

1

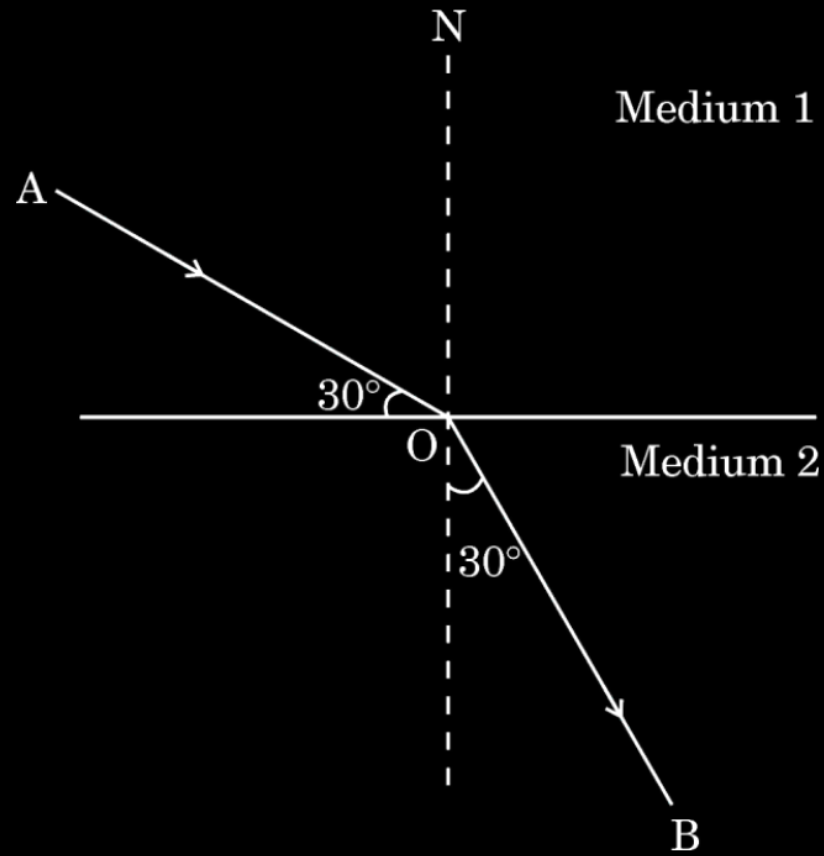
- (A) Concave mirror; Nature of image – real
- (B) Concave mirror; Nature of image – virtual
- (C) Convex mirror; Nature of image – real
- (D) Convex mirror; Nature of image – virtual

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(A) / Concave mirror, Nature of image-real

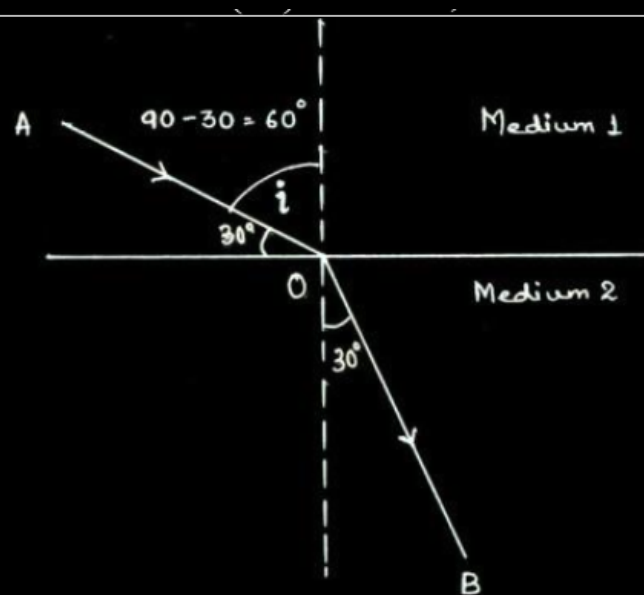
33.

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With the help of the given ray diagram, calculate the refractive index of Medium 2 with respect to Medium 1.

(Given that $\sqrt{3} = 1.732$)



$$\frac{\sin i}{\sin r} = n$$

$$\frac{1}{2}$$

$$\angle i = 90^\circ - 30^\circ = 60^\circ$$

$$\frac{\sin 60^\circ}{\sin 30^\circ} = n$$

$$1$$

$$n = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}$$

$$n = \sqrt{3}$$

$$\frac{1}{2}$$

- 35.** (a) Write the expression for the magnification produced by a lens in terms of object distance and image distance.
- (b) A 4 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 20 cm. Calculate the size of the image formed, if the distance of the object from the lens is 10 cm.

3

- 35.** (a) Write the expression for the magnification produced by a lens in terms of object distance and image distance.
- (b) A 4 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 20 cm. Calculate the size of the image formed, if the distance of the object from the lens is 10 cm.

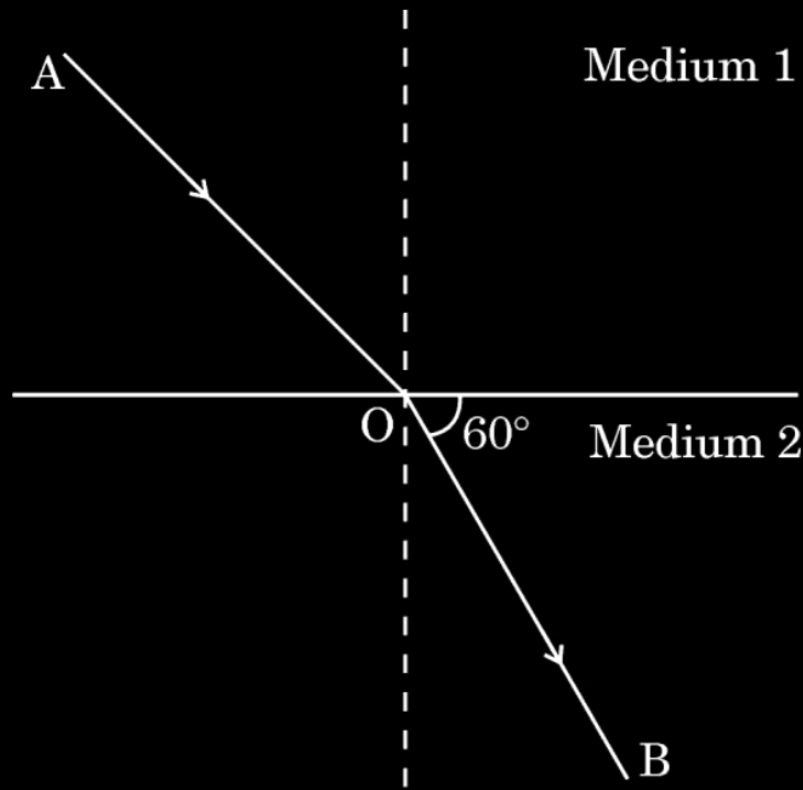
3

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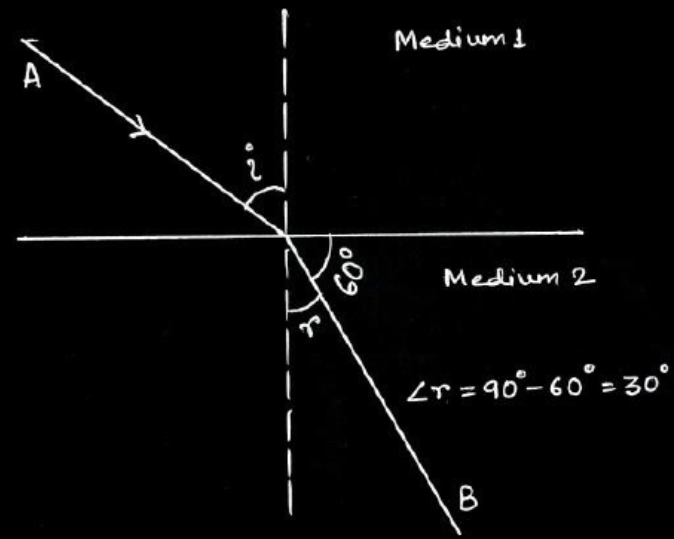
(a) Magnification, $m = \frac{\text{Image distance}}{\text{Object distance}} / m = \frac{v}{u}$	1
(b) $h_o = +4 \text{ cm}$ $f = +20 \text{ cm}$ $u = -10 \text{ cm}$ $h_i = ?$ $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$ $\frac{1}{v} = \frac{1}{20} + \frac{1}{-10}$ $\frac{1}{v} = \frac{-1}{20}$ $v = -20 \text{ cm}$ $m = \frac{v}{u}$ $m = \frac{-20}{-10}$ $m = 2$ $m = \frac{\text{height of image}}{\text{height of object}}$ $\text{height of image} = m \times \text{height of object}$ $\text{height of image} = 2 \times 4 \text{ cm} = 8 \text{ cm}$	$\frac{1}{2}$ 1 $\frac{1}{2}$

33.

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Use the above given ray diagram to calculate the angle of incidence. Given that the refractive index of Medium 2 with respect to the Medium 1 is $\sqrt{3}$.



$$\frac{\sin i}{\sin r} = n$$

$$\frac{1}{2}$$

$$\angle r = 90^\circ - 60^\circ = 30^\circ$$

$$\sin i = n \sin r$$

$$\sin i = \sqrt{3} \sin 30^\circ$$

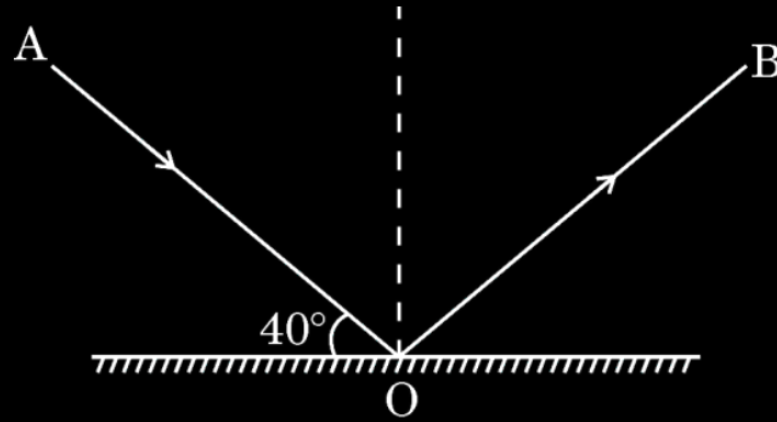
$$1$$

$$\sin i = \sqrt{3} \times \frac{1}{2}$$

$$i = 60^\circ$$

$$\frac{1}{2}$$

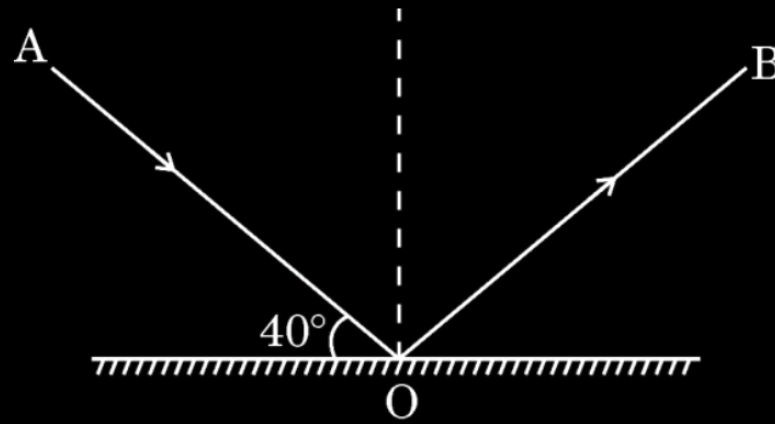
35.



- (a) State the laws which help to find out the angle of incidence and angle of reflection in the ray diagram given above.
- (b) Find out the angle of incidence and angle of reflection in the given ray diagram.

35.

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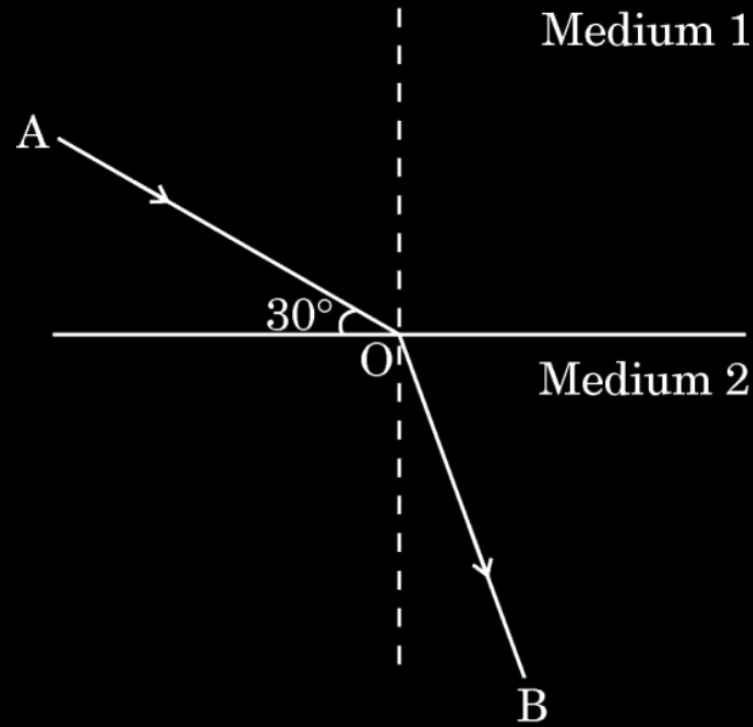
- (a) State the laws which help to find out the angle of incidence and angle of reflection in the ray diagram given above.
- (b) Find out the angle of incidence and angle of reflection in the given ray diagram.

3

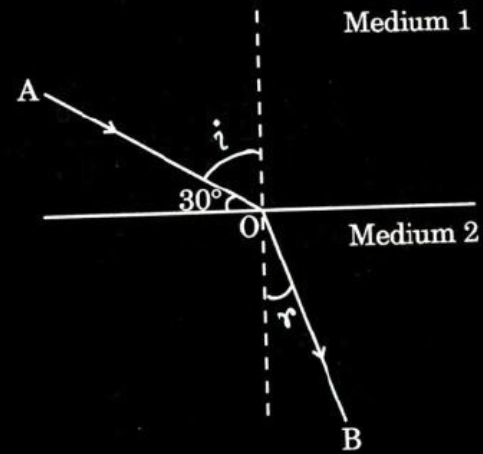
(a)	
• Angle of incidence is equal to angle of reflection.	1
• The incident ray, the normal to the mirror at the point of incidence and the reflected ray, all lie in the same plane.	1
(b) $\angle i = 90^\circ - 40^\circ = 50^\circ$ $\angle r = \angle i = 50^\circ$	$\frac{1}{2}$ $\frac{1}{2}$

33.

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With the help of above given ray diagram, calculate the angle of refraction. Given that the refractive index of Medium 2 with respect to Medium 1 is $\sqrt{3}$.



$$\frac{\sin i}{\sin r} = n$$

$$\angle i = 90^\circ - 30^\circ = 60^\circ$$

$$\sin r = \frac{\sin i}{n}$$

$$\sin r = \frac{\sin 60}{\sqrt{3}}$$

$$\sin r = \frac{\sqrt{3}}{2\sqrt{3}}$$

$$\sin r = \frac{1}{2} \Rightarrow \sin r = \sin 30$$

$$r = 30^\circ$$

$$\frac{1}{2}$$

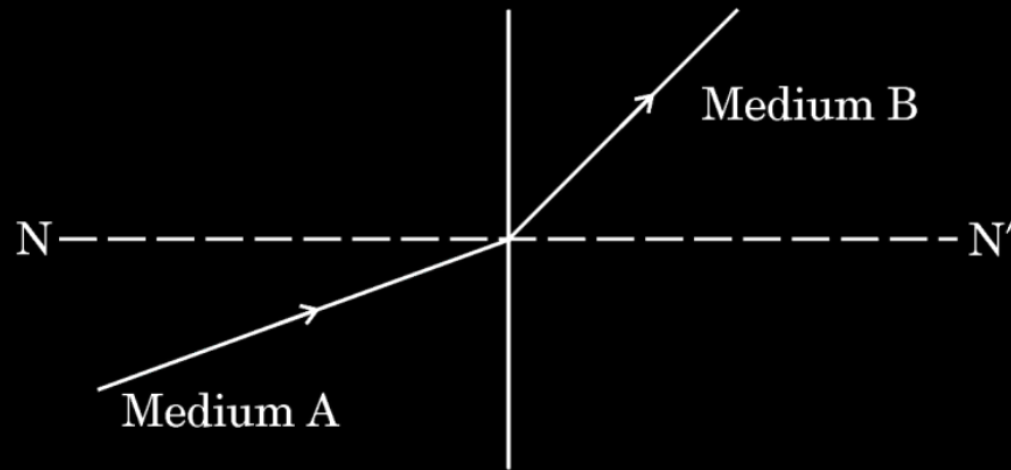
$$1$$

$$\frac{1}{2}$$

- 35.** (a) Write the name of the lens and position of the object in front of it so that a virtual and magnified image is formed.
- (b) An object is placed perpendicular to the principal axis of a convex lens of focal length 20 cm. If object is located at 30 cm from this lens, using lens formula find out the position and nature of the image formed.

33.

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A ray of light enters into medium B from medium A as shown in the given figure :

- (a) Which one of these two media is optically denser than the other medium ? Justify your answer.
- (b) If the speed of light in medium A is V_A and in medium B is V_B , then what is the refractive index of medium B with respect to medium A ?

38. The variation of image distance (v) with the object distance (u) for a convex lens is given in the following observation table. Analyse it and answer the questions given below :

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S.No.	Object distance (u) cm	Image distance (v) cm
1	-150	$+30$
2	-75	$+37.5$
3	-50	$+50$
4	-37.5	$+75$
5	-30	$+150$
6	-15	$+37.5$

- (a) Without calculation, find out the focal length of the given convex lens. Justify your answer. 1
- (b) Which one of the observations given in the table is not correct and why ? 1

- (c) (i) Find out the value of magnification for $u = -30$ cm. Write the nature of the image formed. Give reason for your answer.

2

OR

- (c) (ii) Analyse the data given in the table above and draw the labelled ray diagram for $u = -30$ cm and $v = 150$ cm.

2

(a) In observation no.3. The object distance is equal to image distance, which is only possible when the object is at $2F$ (C) = 50 cm. Therefore, the focal length is 25 cm.

1

(b)

- Observation no. 6
- Because when object is kept between F and O , image formed by lens will be on same side of object. Hence, image distance(v) should be negative.

$\frac{1}{2}$

$\frac{1}{2}$

(c) (i) $m = \frac{v}{u}$
 $= \frac{150}{-30}$
 $m = -5$

$\frac{1}{2}$

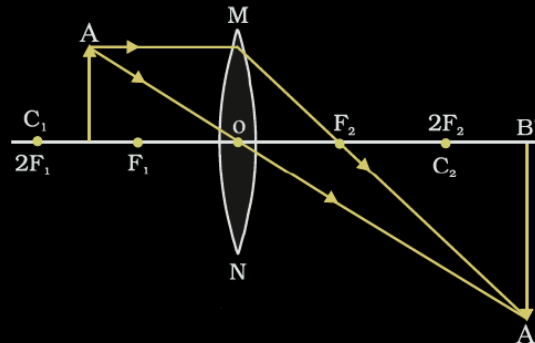
$\frac{1}{2}$

- Nature of image: Image will be real and inverted
- Reason: because the value of m is negative.

$\frac{1}{2}$

$\frac{1}{2}$

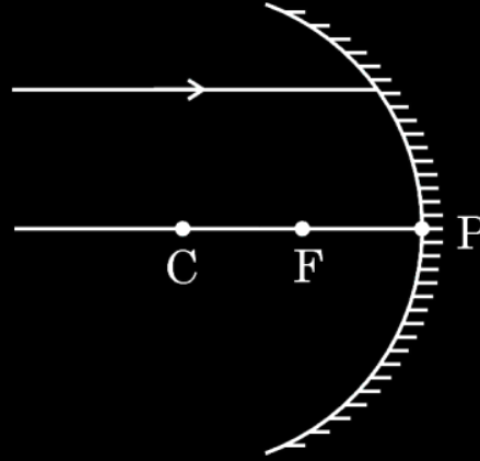
(c) (ii)



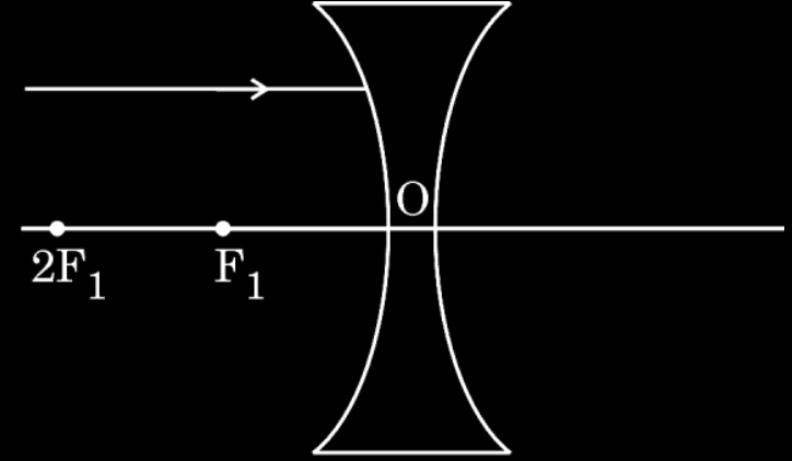
2

34. Redraw the following ray diagrams in your answer book and show the path of reflected/refracted ray :

2



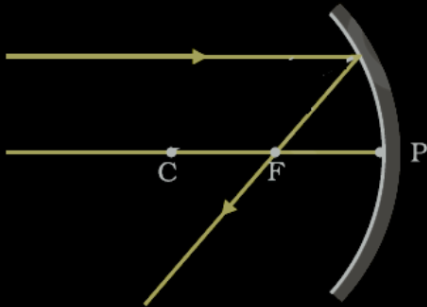
(i)



(ii)

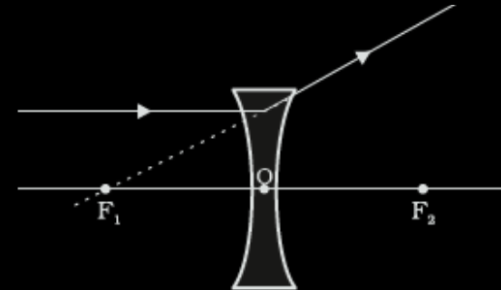
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(i)



1

(ii)



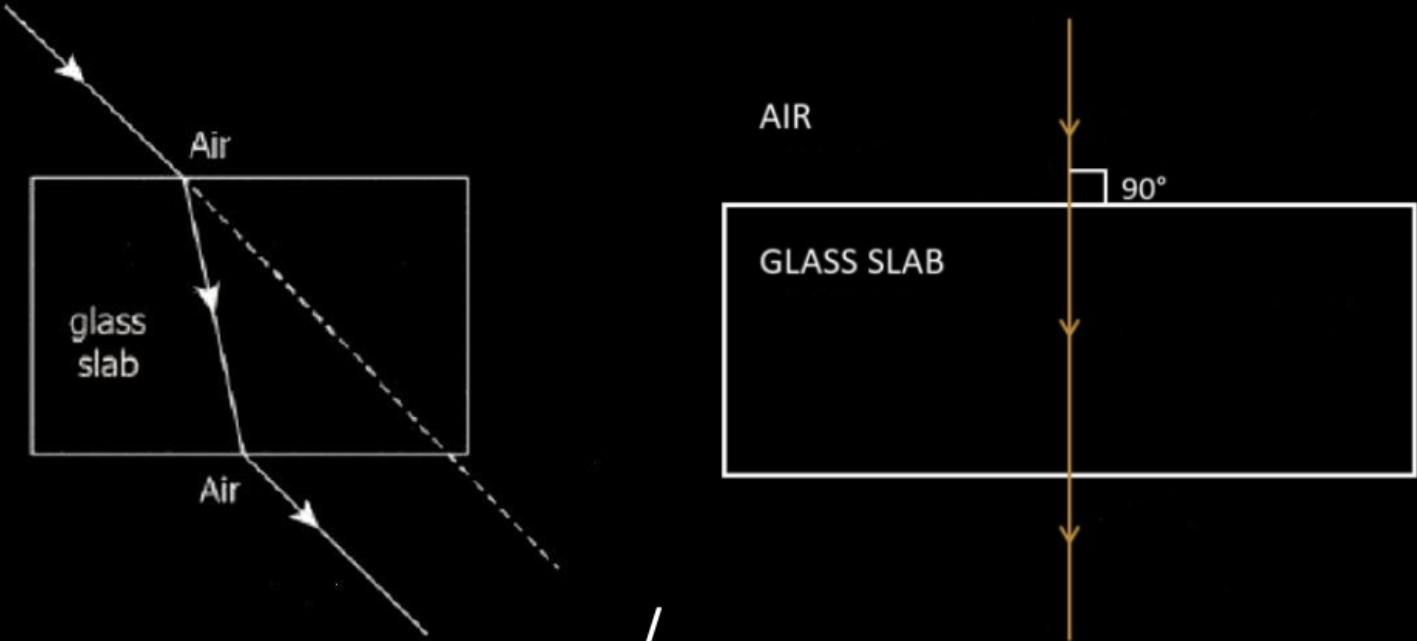
1

- 33.** Draw the ray diagram showing refraction through a glass slab and write the relation between angle of incidence, angle of refraction and refractive index of glass.

33. Draw the ray diagram showing refraction through a glass slab and write the relation between angle of incidence, angle of refraction and refractive index of glass.

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2

 • $\text{Refractive index} = \frac{\sin i}{\sin r}$	1 1
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For Question number 32, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of Assertion (A).
- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Assertion (A) is false, but Reason (R) is true.

32. *Assertion (A) :* When light passes from one medium to another medium, the speed of light may increase or decrease.

Reason (R) : Refractive index of one medium with respect to the other medium may be greater or less than 1.

For Question number 32, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
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- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Assertion (A) is false, but Reason (R) is true.

32. *Assertion (A) :* When light passes from one medium to another medium, the speed of light may increase or decrease.

Reason (R) : Refractive index of one medium with respect to the other medium may be greater or less than 1.

1

(A) / Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

33. Write any two differences between the images formed by convex mirror and plane mirror.

2

33. Write any two differences between the images formed by convex mirror and plane mirror.

2

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	Images formed by convex mirror	Images formed by plane mirror
1	Size of image is always smaller than object.	Size of image is same as size of object.
2	Image distance is not equal to object distance.	Image distance is equal to object distance.

1

1

(any other suitable difference)

(b) When a ray of light passes from water to air (I) will the angle of refraction in air (r) be greater than the angle of incidence in water (i) or less than it ? (II) and when we increase the angle of incidence in water, will the angle of refraction in air increase or decrease ? What is the limiting value for $\angle r$?

2

(b) When a ray of light passes from water to air (I) will the angle of refraction in air (r) be greater than the angle of incidence in water (i) or less than it ? (II) and when we increase the angle of incidence in water, will the angle of refraction in air increase or decrease ? What is the limiting value for $\angle r$?

2

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(b)	(I)	$\angle r$ will be greater than $\angle i$.	1
	(II)		
	•	$\angle r$ will increase.	$\frac{1}{2}$
	•	Limiting value of $\angle r$ will be 90° .	$\frac{1}{2}$

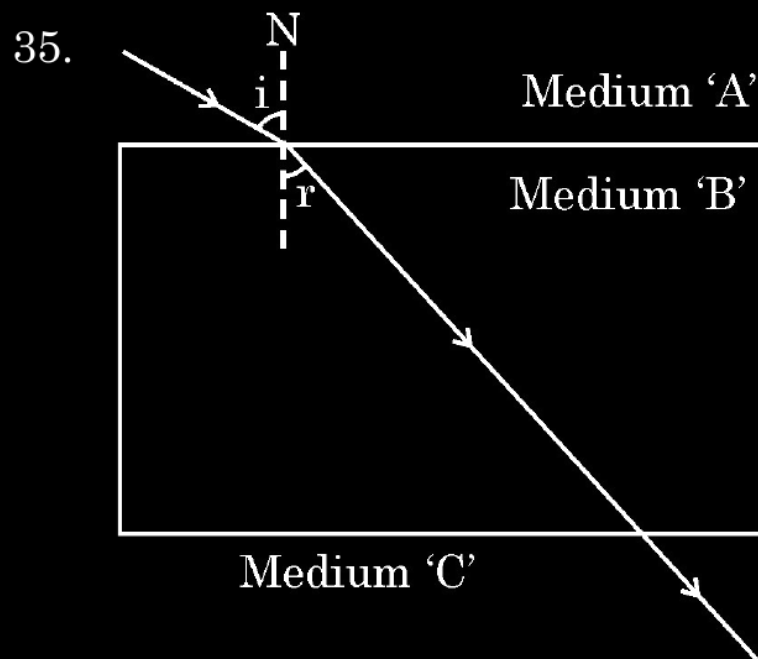
36.

Optical device	Object distance (cm)	Focal length (cm)	Height of object (cm)
Convex lens	20	10	6
Concave mirror	30	10	6

By using the data given in the table, compare the properties of images formed by convex lens and concave mirror in terms of the positions and nature of images formed.

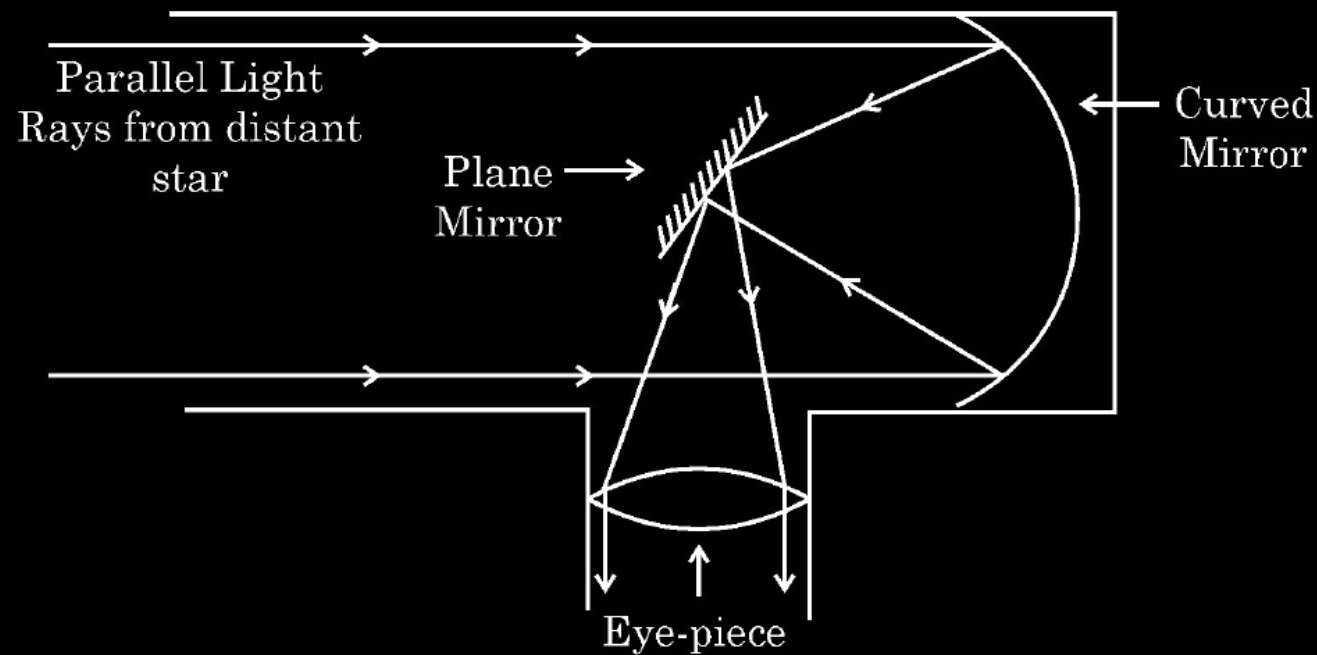
For Convex lens - Position: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$ $\frac{1}{v} = \frac{1}{10} + \frac{1}{-20}$ $v = +20cm$ - Nature: real, inverted	1 ½
For Concave mirror - Position: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$ $\frac{1}{v} = \frac{1}{-10} - \frac{1}{-30}$ $v = -15cm$ - Nature : real, inverted	1 ½

Alternate answer Convex lens: - - When object is placed at centre of curvature, the image is formed at centre of curvature. - Nature of image is real and inverted. Concave Mirror: - - When object is between infinity and centre of curvature the image is formed between focus and centre of curvature. - Nature of image is real and inverted.	1 ½ 1 ½
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- (a) Observe the given ray diagram and compare :
- (i) the speed of light in medium A and the speed of light in medium B.
 - (ii) the optical densities of medium B and medium C.
- (b) Draw a ray diagram for refraction where the optical density of medium 'A' becomes equal to the optical density of medium 'C' and optical density of medium 'B' is greater than the optical density of medium 'A' and medium 'C'.

38.



The above ray diagram is to show image formation by a reflecting telescope. Reflecting telescope revolutionized our way of looking into the sky. Reflecting telescopes employ large mirrors to gather and focus light rather than relying solely on lenses as we do in the refracting telescopes. These telescopes utilize precisely shaped and polished mirrors to capture incoming light and reflect it to a focal point, where the telescope forms an image for observation.

- (a) What kind of image of a star is seen by the observer in the eye-piece ?
- (b) What kind of mirror is used in this reflecting telescope ?

- (c) Explain with the reason, what kind of optical device (type of lens or mirror) is used at the eye-piece.

OR

- (c) What is the role of plane mirror in image formation by this telescope ?

- (c) Explain with the reason, what kind of optical device (type of lens or mirror) is used at the eye-piece.

OR

- (c) What is the role of plane mirror in image formation by this telescope ?

(a) Real / magnified.	1
(b) Converging / Concave mirror, Plane mirror	1
(c)	
• Convex lens	1
• It magnifies the image formed by the curved mirror	1
OR	
(c) The plane mirror redirects/ reflects the light rays coming from the curved mirror towards eyepiece.	2

37. A mirror used for rear-view of an automobile has a focal length 2.0 m. If a car is located at 8.0 m from the mirror - 3

(a) Find out the position of the image of this car as seen in the mirror.

(b) Draw the ray-diagram for image formation by the rear-view mirror. CBSE 26

37. A mirror used for rear-view of an automobile has a focal length 2.0 m. If a car is located at 8.0 m from the mirror - 3

- (a) Find out the position of the image of this car as seen in the mirror.
 (b) Draw the ray-diagram for image formation by the rear-view mirror. CBSE 26

(a) $f = +2\text{m}$ $u = -8\text{m}$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

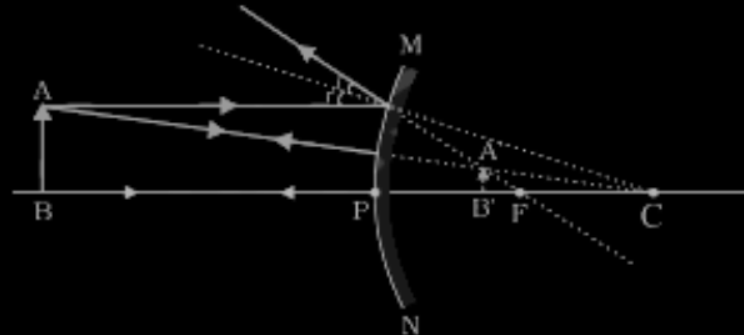
$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{2} + \frac{1}{8}$$

$$v = \frac{8}{5} = 1.6\text{ m}$$

The image is formed 1.6m behind the mirror

(b)



$\frac{1}{2}$

$\frac{1}{2}$

1

1

30. Rays from the sun converge at a point 25 cm behind a convex lens. The distance at which an object be placed in front of the lens to get a virtual image, is :

(A) 20 cm

(B) 40 cm

(C) 50 cm

(D) More than 50 cm

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30. Rays from the sun converge at a point 25 cm behind a convex lens. The distance at which an object be placed in front of the lens to get a virtual image, is :

1

(A) 20 cm

(B) 40 cm

(C) 50 cm

(D) More than 50 cm

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(A) / 20 cm

33. (a) Relate the speed of light in the given medium with its optical density.
- (b) Using the information given in the table below, arrange the medium A, B and C in the ascending order of their optical density.

Medium	Speed of light (m/s)
A	2.25×10^8
B	2×10^8
C	2.08×10^8

33. (a) Relate the speed of light in the given medium with its optical density.
- (b) Using the information given in the table below, arrange the medium A, B and C in the ascending order of their optical density.

Medium	Speed of light (m/s)
A	2.25×10^8
B	2×10^8
C	2.08×10^8

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(a) In optically rarer medium speed of light is more and in optically denser medium speed of light is less.	1
(b) $A < C < B$	1

36. (a) A ray of light after reflection from a concave mirror becomes parallel to its principal axis. Draw a labelled ray diagram to show this. 3
- (b) (i) Differentiate between the virtual images formed by a convex mirror and concave mirror.
- (ii) “The value of magnification ‘ m ’ for a mirror is -2 .” Calculate the height of object if the height of image is 20 cm. CBSE 26

36. (a) A ray of light after reflection from a concave mirror becomes parallel to its principal axis. Draw a labelled ray diagram to show this.

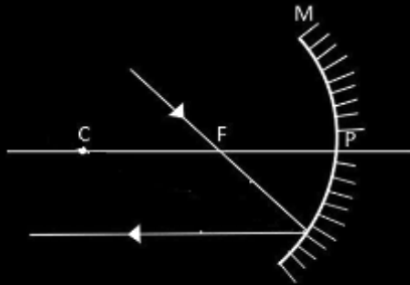
3

(b) (i) Differentiate between the virtual images formed by a convex mirror and concave mirror.

(ii) “The value of magnification ‘ m ’ for a mirror is -2 .” Calculate the height of object if the height of image is 20 cm.

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(a)



(b)

(i)

Virtual Image formed by Convex mirror	Virtual Image formed by Concave mirror
Image is formed for all positions of the object	Image is formed when object is placed between Pole and Principal Focus
Small/diminished	Enlarged/magnified

(Any one difference)

1

1

(ii) $m = -2$

$$m = \frac{\text{height of image}}{\text{height of object}} = \frac{h'}{h} \quad \frac{1}{2}$$

$$h = \frac{h'}{m}$$

$$h = \frac{-20}{-2}$$

$$h = 10 \text{ cm}$$

$\frac{1}{2}$

33. (a) Define absolute refractive index of an optical medium.

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2

(b)

Material Medium	Refractive Index
A	1.50
B	1.46
C	1.31
D	1.77

Arrange these material mediums given in the table in increasing order of speed of light through them.

33. (a) Define absolute refractive index of an optical medium.

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2

(b)

Material Medium	Refractive Index
A	1.50
B	1.46
C	1.31
D	1.77

Arrange these material mediums given in the table in increasing order of speed of light through them.

(a) Absolute refractive index of a medium is the ratio of speed of light in air to the speed of light in the given medium / Refractive index of a medium with respect to air /	1
$n = \frac{c}{v}$	
(b) $V_D < V_A < V_B < V_C$	1

36. An object is placed at a distance of 30 cm in front of a convex lens of focal length 15 cm. Use lens formula to determine the position of the image. What will be the size of the image in this case ?

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36. An object is placed at a distance of 30 cm in front of a convex lens of focal length 15 cm. Use lens formula to determine the position of the image. What will be the size of the image in this case ?

3

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- $u = -30 \text{ cm}, f = 15 \text{ cm}$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{15} + \frac{1}{-30}$$

$$v = 30 \text{ cm}$$

$$\frac{1}{2}$$

- $m = \frac{v}{u} = + \frac{30}{-30} = -1$

Hence, size of image will be same as the object.

/

Size of image will be same as size of object because when object is kept at $2F_1$, image will be formed at $2F_2$.

$$\frac{1}{2}$$

$$1$$

$$1$$

33. When an incident ray of light enters in a medium 'X' from medium 'Y', it bends away from the normal.

2

Comment about the following :

- (a) Speed of light in medium 'X' with respect to the speed of light in medium of 'Y'.
- (b) Optical density of medium 'X' with respect to the optical density of medium 'Y'.

Give reason for your answer in each case.

33. When an incident ray of light enters in a medium 'X' from medium 'Y', it bends away from the normal.

2

Comment about the following :

- (a) Speed of light in medium 'X' with respect to the speed of light in medium of 'Y'.
- (b) Optical density of medium 'X' with respect to the optical density of medium 'Y'.

Give reason for your answer in each case.

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- | | | |
|-----|--|---------------|
| (a) | <ul style="list-style-type: none">• Speed of light in medium 'X' is more than in medium 'Y'.• Because light ray bends away from the normal in optically rarer medium. | $\frac{1}{2}$ |
| (b) | <ul style="list-style-type: none">• Optical density of medium 'X' is less than medium 'Y'.• Because in optically rarer medium light bends away from the normal. | $\frac{1}{2}$ |

36. An object of height 6 cm is placed at a distance of 30 cm from the optical centre of a concave lens of focal length 15 cm.

3

Use lens formula to determine :

- (a) The distance of image from the optical centre.
(b) The height of the image formed.

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(a)	Height of object = 6.0 cm $u = -30$ cm $f = -15$ cm $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$ $\frac{1}{v} = \frac{1}{-15} + \frac{1}{-30}$	$\frac{1}{2}$ $\frac{1}{2}$
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(b)	$v = -10$ cm $m = \frac{v}{u}$ $m = \frac{-10}{-30}$ $m = \frac{1}{3}$ $m = \frac{\text{height of image}}{\text{height of object}} = \frac{h'}{h}$ $h' = m \times h$ $h = \frac{1}{3} \times 6$ $h = 2$ cm	1 $\frac{1}{2}$ $\frac{1}{2}$
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